

first name \_\_\_\_\_ last name \_\_\_\_\_ email \_\_\_\_\_  
In all quizzes and exams, show your work and write with correct notation.

(10pts) 1. Find  $\lim_{x \rightarrow 3} \frac{x^2 - 7x + 12}{x - 3}$

$\lim_{x \rightarrow 3} \frac{(x-3)(x-4)}{x-3}$

$\lim_{x \rightarrow 3} (x-4)$

$= 3 - 4$

$= -1$

Because  $\lim_{x \rightarrow 3} (x-3) = 0$   
we can't split.

No limit notation here.

No limit notation here.

(10pts) 2. Find  $\lim_{x \rightarrow 6} \frac{x-6}{x^2-4x-12}$

$\lim_{x \rightarrow 6} \frac{x-6}{(x-6)(x+2)}$

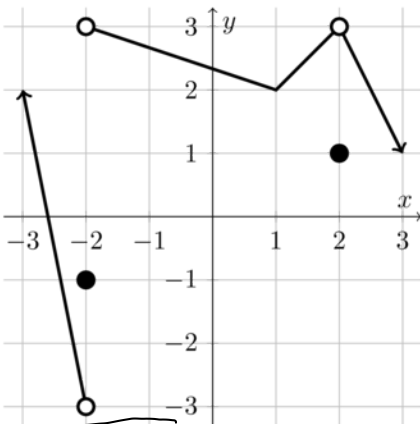
$\lim_{x \rightarrow 6} \frac{1}{x+2}$

$\lim_{x \rightarrow 6} \frac{1}{8}$

$\lim_{x \rightarrow 6} (x+2) = 8$

Notice all the equal signs.

(10pts) 3. Based on the graph of  $y = f(x)$  below, find  $8 \sin\left(\frac{5\pi}{6}\right) + \lim_{x \rightarrow 2} f(x)$



$\lim_{x \rightarrow 2} f(x) \neq 1$   
 $f(2) = 1$

Be careful to not confuse  $\lim_{x \rightarrow 2} f(x)$  and  $f(2)$   
not equal for this function

$= 8 \cdot \frac{1}{2} + 3$   
 $= 4 + 3$   
 $= 7$

Note: You are expected to know trig function values at the special angles on the unit circle. If you want to understand the unit circle (instead of memorizing and forgetting) please talk to me in student hours.

